

ESTIMATION OF INTERNAL REPRESENTATION OF VOWELS IN COCHLEAR IMPLANT USERS IN RELATION WITH FREQUENCY-TO-PLACE MISMATCH & NEURAL RESPONSIVESNESS

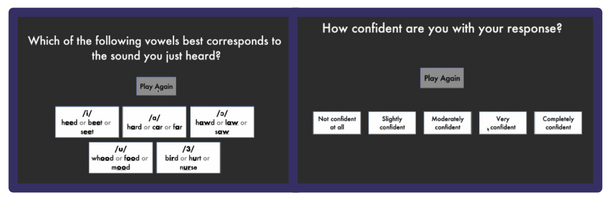
L. Villejoubert [1], L. Picinali [2], K. Faulkner [3], D. Vickers [1]

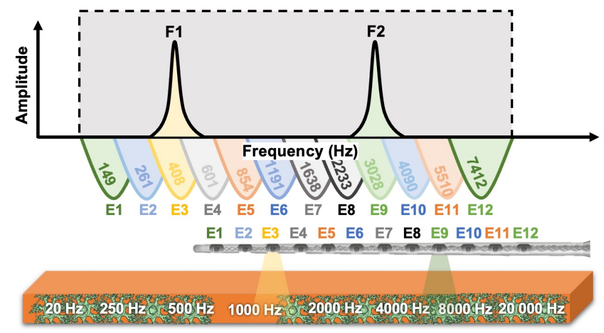
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The present study is part of a larger project aimed at providing clinicians with a testing solution to evaluate how cochlear-implant (CI) users adapt to their devices in relation with specific individual factors.

Some CI users remain dissatisfied with their perception, especially in challenging listening situations. Here, we evaluate differences in the perception of a speech component - namely, vowels - in adults who have extensive experience with their devices.

In theory, their vowel internal representations should not differ substantially from those of their normal-hearing (NH) peers; this is precisely what we investigate in the first part. In the second part of the poster, we explore whether differences between CI users might be explained by some specific individual factors explored here.

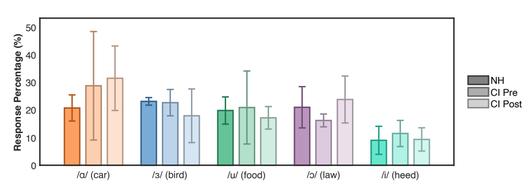




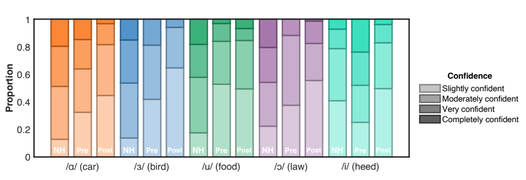
GOAL 1 — **Do NH listeners' and CI users' perceptions differ in their internal vowel representations?**

GOAL 2 — **Understand whether CI users' perception is influenced by frequency-to-place factors**

- NORMAL HEARING LISTENERS VS COCHLEAR IMPLANT USERS -



Comparison of response proportions among the 3 groups tested



Comparison of response proportions among different response degrees for each vowel and across the different groups

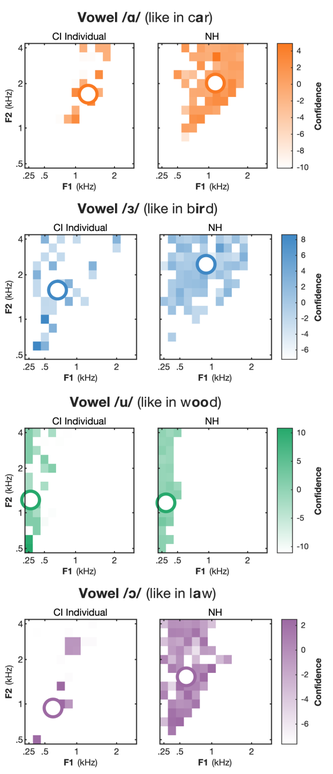
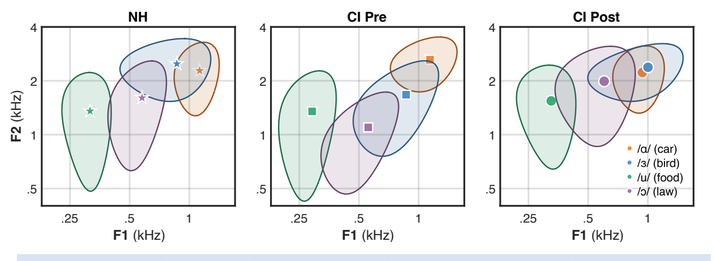
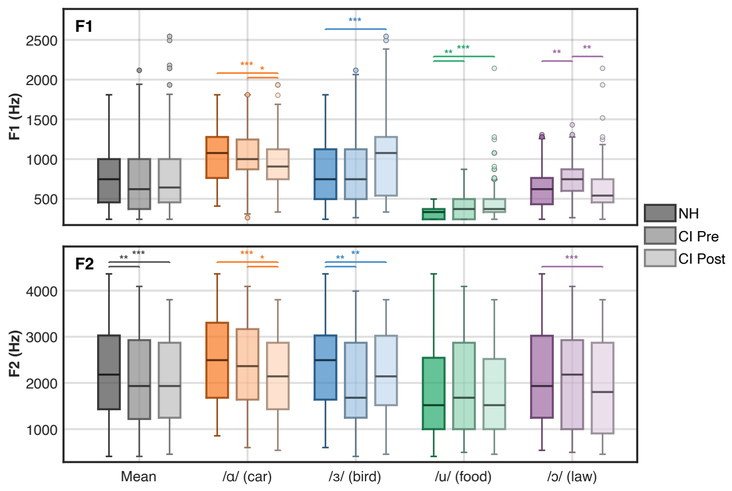


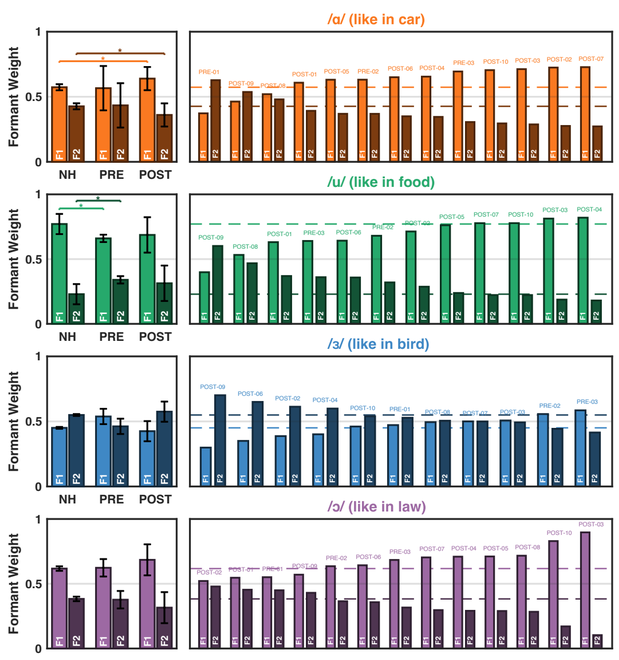
Illustration of response data for one CI user for the 4 vowels kept in the analysis. For each vowel, the left panel represents the CI user's individual data and the right panel the mean response for NH listeners. The color intensity represents the confidence scale. The data presented here is the output of the Bayesian model. The circle represents the weighted mean (i.e., mean computed using the confidence scale). This is then used to calculate the F1 and F2 shifts for each vowel and each CI user. The shift is calculated using the weighted mean of NH listeners (right panels).



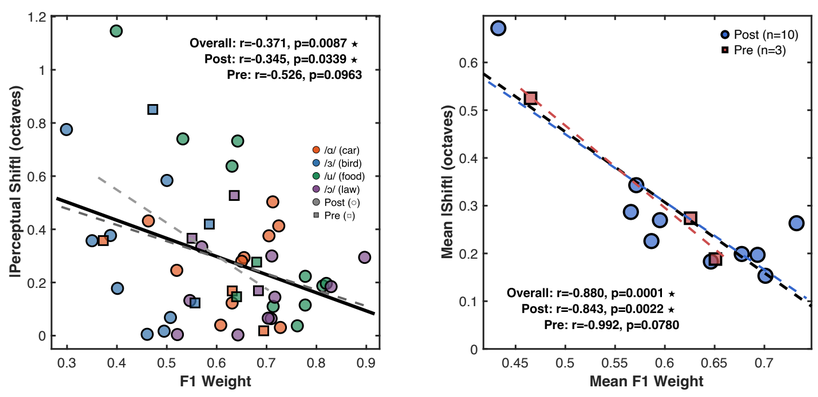
Views representing the responses of the participants for the 3 testing groups considered in this experiment. The x-axis corresponds to the F1 frequency and the y-axis to the F2 frequency. The markers represent the mean responses, and the shaded areas represent the confidence ellipses. Each color represents one vowel.



Comparison of responses among the 3 groups, with F1 and F2 responses separated and statistics added. Results indicate a general trend toward higher F1 and lower F2 for CI users compared to their NH peers. This might indicate a compressed perception of vowel identification for CI users.

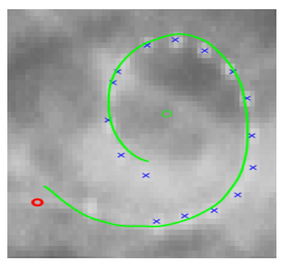


Calculation of formant importance for each group and each CI user independently. The calculation was done using the weighted spread of responses. Concretely, if the response of a participant for a specific vowel and formant is consistent (*i.e.*, a very low spread in frequency), the importance of this formant would be high. It is worth noting that the importance is always a ratio between F1 and F2. The absolute ratio for CI users is interesting to analyse in comparison to the NH ratio.

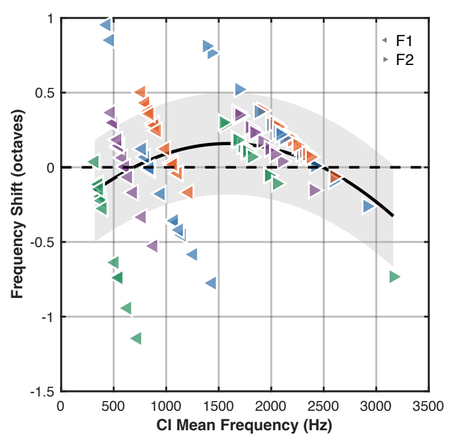


The plots above illustrate the relationship between perceptual shift (or vowel shift) and formant weight (illustrated here with F1 Weight). There is a clear correlation between the two, with greater shift associated with lower reliance on F1.

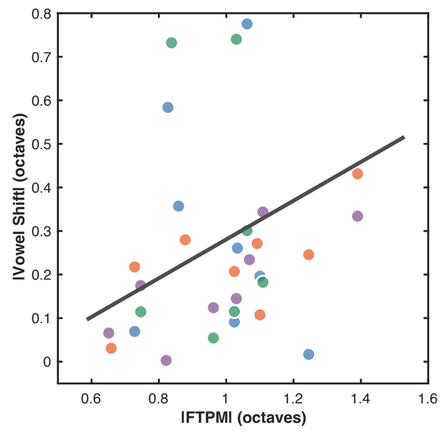
EFFECT OF FTPM



To evaluate a possible effect of FTPM (also referred to here as "physical" shift, as opposed to "perceptual" shift for vowel shift), we calculated the **actual insertion depth** of each electrode using an algorithm developed by Dr. Iwan Roberts (from the Bance laboratory at the University of Cambridge), using x-ray images. We then compared these insertion depth values with the Stakhovskaya formula, which is used to calculate the corresponding **characteristic frequency of the spiral ganglion cells** for each electrode. Finally, these characteristic frequencies were used to **calculate FTPM by comparing them with the allocated frequencies**. Perceptual and physical shifts were then compared.

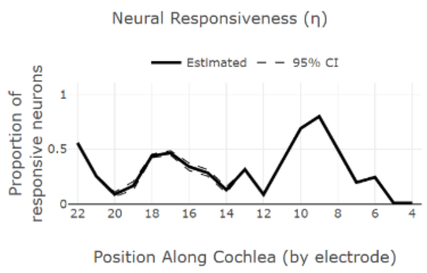


Representation of the frequency shift for each vowel's F1 and F2 for all groups. A negative shift means that the shift is toward high frequencies. In this plot, no information about the "physical" shift (or FTPM) is represented. The usefulness of this representation is that it illustrates the shape of the shift, with a shift that appears to be non-linear along the electrode array: the perceptual shift is more pronounced toward high frequencies for low electrodes, then a shift toward low frequencies for frequencies between 1000 and 2500 Hz, and then back toward high frequencies for higher frequencies. This shape might resemble the shape of physical FTPM, which shows more shift at low and high frequencies and a reduced one for mid-frequencies.

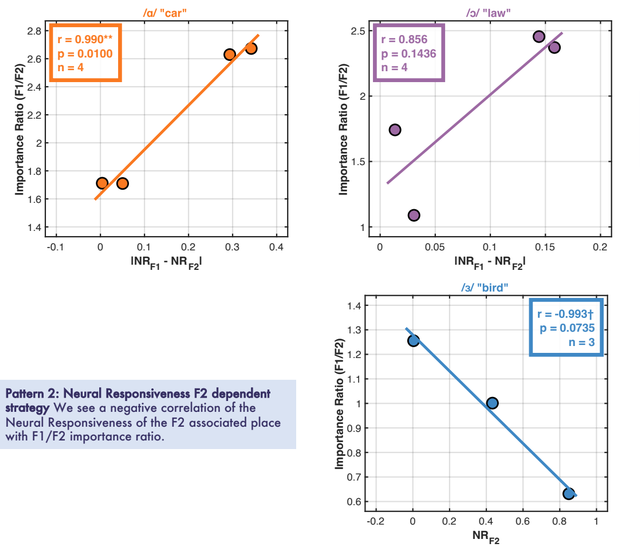


This plot illustrates the relationship between vowel shift and FTPM. Data reported here are in absolute values, because otherwise the perceptual and physical shifts would be in opposite directions (most of the time), which might be explained by a compensation process.
A possible "incomplete" adaptation is seen here in the ratio/proportion between Vowel Shift and FTPM scales: the trend here is probably a mark of incomplete adaptation toward the physical shift. What is illustrated here is the fact that the greater the physical shift, the greater the perceptual shift.

EFFECT OF NEURAL RESPONSIVENESS



Another assumption explored here is that the quality of the ENI should impact the identification process. This was based on the fact that, in the case of degraded ENI, the quality of the signal would be impacted and probably distorted. Many analyses were conducted, and most of them were not significant or did not show any trend between the values collected using the PECAP method (Garcia et al., 2020) and the behavioural responses. The data presented here represent the first trend that we observed between neural responsiveness and behavioural responses.
We found 2 patterns illustrated below.



Pattern 1: Imbalance-Driven Strategy
The absolute difference between Neural Responsiveness F2 and F1 - regardless of direction - strongly predicts F1 importance

Pattern 2: Neural Responsiveness F2 dependent strategy We see a negative correlation of the Neural Responsiveness of the F2 associated place with F1/F2 importance ratio.